R Worksheet: Matrices, Plots, Libraries, and Functions

Section 1: Matrices + Operations

Three students take a class together and study for a **project**, **midterm**, and **final** together.

- Student A scores: `12, 23, 34`
- Student B scores: `45, 56, 67`
- Student C scores: `78, 89, 64`

Q1: Create a matrix containing their scores.
Use `matrix()` to create the 3x3 matrix:

```r
# Solution:
m <- matrix(
  c(12, 45, 78,
    23, 56, 89,
    34, 67, 64),
  nrow = 3,
  ncol = 3,
  byrow = TRUE
)
```

Q2: Name the rows and columns
Give descriptive names to the columns (assignments) and rows (students) using `colnames()` and `rownames()`.

```r
# Solution
colnames(m) <- c("Project", "Midterm", "Final")
rownames(m) <- c("Student A", "Student B", "Student C")
```

Q3: Compute each student's average score
Which student has the highest average?

```r
# Solution
rowMeans(m)
```

Plots

Q4: How can we make simple graphs in Base R?
Solution:

Use `hist()`, `plot()`, and `boxplot()` to visualize data.

Q5: Create a histogram using the matrix scores
Hint: You can use `as.vector(m)` to convert matrix to a vector.

```r
# Solution:
hist(as.vector(m))
```

Q6: Use the built-in `airquality` dataset

Let's use the airquality dataset, which has daily New York air quality measurements (May–Sept 1973). Variables include:

- `Ozone`: Ozone concentration (ppb)
- `Solar.R`: Solar radiation (langley)
- `Wind`: Wind speed (mph)
- `Temp`: Temperature (°F)
- `Month`: Month (5 = May, …, 9 = September)
- `Day`: Day of the month

a) Histogram of Ozone (Hint: use $ to access a column from a data frame)

```
# Solution:

hist(airquality$Ozone)
```

b) Scatterplot: Ozone vs Temperature

```
# Solution:

plot(airquality$Temp, airquality$Ozone,
     xlab = "Temperature (°F)",
     ylab = "Ozone (ppb)",
     main = "Ozone vs Temperature")
```

c) Boxplot: Temperature by Month

```
# Solution:

boxplot(Temp ~ Month, data = airquality,
        xlab = "Month",
        ylab = "Temperature (F)",
        main = "Temperature by Month")
```

Intro to Libraries

Q7: Install and load the libraries

```
install.packages("igraph")
install.packages("networkD3")

library(igraph)
library(networkD3)
```

Q8: What are `igraph` and `networkD3`?

How are they useful for data analysis and visualization?

Write lines of code to at the documentation:

```
# Solution:

?igraph
?networkD3
```

Q9: Explore usage

- `igraph` is used for creating and analyzing network graphs.
- `networkD3` helps build interactive network visualizations.

Run these examples on your own computer to understand each package's usage.

Example using `igraph`:

```
g <- graph(edges = c(1,2, 2,3, 3,1), n = 3, directed = FALSE)
plot(g)
```

Example using `networkD3`:

```
simpleNetwork(data.frame(source = c("A", "B", "C"),
                         target = c("B", "C", "A")))
```

Functions

Q10: Write a function to simulate rolling a die without replacement

```
# Solution:

roll_die <- function(n = 6) {
  sample(1:6, n, replace = FALSE)
}

roll_die()
```